SREE LASYA BALAJI

Boston, MA 02120 | +1 (857) 308-6305 | balaji.sree@northeastern.edu | LinkedIn | Portfolio

Available: January 2027 for Co-op or Internship

Education

MS in Computer Science | Northeastern University | Boston, MA *Jan 2026 – Expected May 2028*
Courses: Database Management Systems, Machine Learning, AI Foundations **GPA: 3.6/4.0**

B.Tech in Computer Science & Engineering | Sreenidhi Institute | Hyderabad, India *Jun 2025*
GPA: 3.0/4.0

Technical Skills

Programming Languages: Python, SQL, JavaScript, HTML, CSS,
Data Engineering: Snowflake, dbt, Databricks, Spark/PySpark, ETL/ELT Pipelines
Business Intelligence: Power BI, Tableau, Microsoft Excel (VLOOKUPs, Pivot Tables), Dashboard Development
Data Practices: Data Validation, Data Quality Assessment, Exploratory Data Analysis, Data Cleaning & Preparation
Database Technologies: MySQL, MongoDB, Relational Database & Schema Design
Backend & Frameworks: FastAPI, Django, Streamlit, REST APIs
ML & AI (Applied): scikit-learn, pandas, Numpy Prompt Engineering, Generative AI Evaluation
Cloud & Certifications: AWS Academy Machine Learning Foundations, AWS Academy Data Engineering, AWS Academy Cloud Foundations, AWS Academy Cloud Architecting
Tools & Practices: Git, Linux, Agile/Scrum

Professional Experience

AI Instructional Assistant | CATLR - Northeastern University | Boston, MA *Sep 2026 – Present*

- Partner with faculty leading **AI Curriculum Transformation** at the **D'Amore-McKim School of Business** and **College of Professional Studies** to build AI competencies and integrate **generative AI** into coursework.
- Follow a structured **prototype → test → evaluate → document → scale workflow** for every AI tool, using **Claude.ai** and other generative AI platforms to engineer reliable, reusable prompting strategies for real **course-assignment use cases**.
- Built a **repeatable, documented evaluation process** to validate AI tool outputs for **accuracy, safety, and reliability**, establishing clear criteria before any tool is recommended for real-world classroom use.
- Research how **peer institutions** and professions approach AI use in teaching, translating findings into **curriculum recommendations** for faculty stakeholders.
- Partner directly with **non-technical faculty stakeholders** to scope requirements and translate them into working, validated solutions, managing **multiple concurrent projects and deadlines**.

Projects

University Budget Management System (UBMS) - Data Pipeline & Full-Stack Application

- Designed a **23-table relational schema** in **MySQL 8**, building **ETL-style data workflows** with **5 stored procedures, 4 user-defined functions, 5 triggers, and 2 scheduled events** to automate processing and enforce data integrity across a university's complete budget operations.
- Built a **FastAPI** backend with **6 role-based API routers** and REST endpoints, ingesting and transforming data for **5 distinct user roles** (Finance Officer, Admin, Dean, Registrar, Student), each with tailored access and validation rules.
- Built **interactive dashboards and reports** in **Next.js 14/React/TypeScript**, translating the distinct reporting needs of all 5 stakeholder groups into tailored, real-time views of budgets, payroll, and enrollment.
- Documented data schemas, API endpoints, dependencies, and access permissions, creating a reference **data dictionary** for long-term maintainability.

Road Safety Image Enhancement (OpenCV) - Deep Learning

- Developed and trained a **deep learning (CNN-based)** image enhancement pipeline, integrated into a Django web application for real-time **object detection** in rain, haze, and fog conditions.
- Applied **Dark Channel Prior** for haze removal, **guided filtering** for transmission refinement, and **CNN-based parameter tuning** for adaptive color and brightness restoration.
- Evaluated model performance against quantitative diagnostic metrics (**PSNR, SSIM**) to identify failure modes and determine the most reliable method for downstream object detection.

Spam Mail Detection - Machine Learning & NLP

- Conducted **exploratory data analysis** and preprocessed raw text using **pandas**, **NumPy**, and **NLTK** (tokenization, stop-word removal, stemming) to engineer features for model training.
- Implemented and compared **five classification models** (Naïve Bayes, SVM, KNN, Decision Tree, Random Forest) using **scikit-learn**, evaluating each on **accuracy and precision** to identify the most reliable approach.
- Built and deployed a **Streamlit web application** serving the trained model and vectorizer (via **pickle**) for real-time spam classification.